A roton-like minimum in the excitation spectrum of quantum spin ice

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Quantum spin ice hosts an emergent photon at long wavelength, and its compact gauge theory is known to be strongly interacting. We show that in the pure ring-exchange model the lowest collective excitation lies not at small momentum but at the zone-boundary L point. Exact spectroscopy of the local-Ising correlation function $S^{zz}(\boldsymbol{q},\omega)$, resolved eigenstate by eigenstate, on 48-, 64-, 72- and 96-site pyrochlore clusters — the largest a 147-million-dimensional constrained sector — finds at every size a single sharp level at 0.90–1.07K carrying at least half of the longitudinal weight at L, below the lowest bright level at every other accessible momentum. Its location and relative depth are predicted by the ground-state Feynman ratio $f_1(\boldsymbol{q})/S(\boldsymbol{q})$: on clusters whose anisotropy splits the L star into inequivalent classes, the ratio selects in advance which class reconstructs, verified independently at 64 and 96 sites, and on the 72-site cluster it predicts the energies of two soft levels at different momenta to better than one percent. Because both factors of the ratio are ground-state quantities, they are computable far beyond exact diagonalization: sign-free Green's-function Monte Carlo on complete momentum grids of up to 2048 sites, validated digit-for-digit against the 48-site spectra, shows that the zone-boundary minimum of f_1/S sits at L at every size, 20–25% deep, drifting by under two percent from 864 to 2048 sites. Exact plaquette decomposition attributes 71% of the mode's variational relaxation to the one hexagon family the probe leaves unconstrained, a lattice analogue of backflow. We state precisely what remains open: the mode's sharpness is established only up to 96 sites, the Monte Carlo constraining the centroid rather than the poles beyond that; and on the lattice the minimum is over the zone-boundary landscape and the bright spectrum, not a radial helium-style dip. Within those stated limits, the identification of a Feynman roton-like minimum in the emergent-photon spectrum is supported at every point by exact or gated computation.

I. INTRODUCTION

Quantum spin ice realizes a compact $U(1)$ gauge theory inside a frustrated magnet. In the two-in–two-out manifold of the pyrochlore lattice, virtual exchange flips the six spins around an elementary hexagon [1, 2]. The resulting deconfined phase supports a gapless emergent photon together with gapped electric and magnetic charges [3–6]. Because the local moment is the emergent electric field, $S^z = E$, magnetic neutron scattering measures electric-field fluctuations through $S^{zz}(\boldsymbol{q},\omega)$. The long-wavelength photon and its characteristic correlations have been established by gauge theory, quantum Monte Carlo, and constrained-space numerics [2, 7–10] and motivate searches in a growing set of candidate materials [11–19].

The microscopic problem is nevertheless not weakly interacting. Gaussian treatments expand the compact magnetic energy about small flux and give a non-compact Maxwell theory whose neutron response is concentrated in one-photon poles [2, 7, 20, 21]. This description is controlled in the infrared, but several results already show that it becomes incomplete at shorter wavelength. The emergent fine-structure constant is large [10]; semiclassical expansions find strong anharmonic renormalization and, in particular, an interaction-induced splitting of the two transverse photon polarizations at intermediate wave vector [22–24]; and quantum Monte Carlo with analytic continuation finds substantial zone-boundary broadening not captured by Gaussian electrodynamics [25]. Constrained-space exact diagonalization has been pushed to larger clusters than ours — to 96 spins in Ref. [10] — but for a different purpose: extracting ground-state and low-energy parameters of the emergent electrodynamics, for which one does not need the full set of eigenstates and matrix elements. Resolving the operator content of the spectrum is more expensive per spin than locating its lowest levels, and that is the trade we make. Within that more demanding calculation we reach 96 spins — matching the largest constrained-space diagonalization on record — which required abandoning the single-word bit-mask representation that caps such work at 64.

The unresolved question we address is therefore narrower: *how is the longitudinal spectral weight of the microscopic ring model distributed among its finite-energy eigenstates at finite momentum?* Analytic continuation can reveal broad dynamical features but not the individual finite-size poles, matrix elements, and state character needed to answer this question sharply, while previous exact diagonalization was organized around low-energy QED parameters rather than the complete operator-resolved response. We compute the matrix elements of $S^z(\boldsymbol{q})$ directly in the constrained Hilbert space, and we complement the exact spectra with sign-free ground-state Monte Carlo that carries the mechanism's two ingredients — both ground-state quantities — to 2048 sites. The result is a long-wavelength photon that survives as

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expected and a zone-boundary spectrum that is strongly reconstructed. The most prominent feature is the lowest excitation of every diagonalized cluster: a single level at L carrying about half of the longitudinal weight there, below the lowest bright level at every other accessible momentum. We call it *roton-like* in a precisely operational sense, defined and defended in Sec. IX: a finite-momentum minimum of the collective spectrum, produced by the Feynman mechanism — the ground-state ratio f_1/S locates it, predictively — and relaxed by a lattice analogue of backflow. What that terminology does and does not commit us to, including the two statements our data cannot make, is set out there rather than left to connotation.

II. MODEL, OBSERVABLE, AND SCOPE

We study the standard pyrochlore ring-exchange Hamiltonian within the spin-ice manifold [2, 8],

$$H = -K \sum_{p} \left(W_p + W_p^{\dagger} \right), \qquad W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-, \tag{1}$$

summed over contractible elementary hexagons. W_p reverses an alternating six-spin configuration and annihilates a nonflippable one. Thus the kinetic energy rewards coherent resonance among flippable hexagons. In gauge language $W_p = e^{iB_p}$ and Eq. (1) is the compact magnetic energy $-2K \sum_p \cos B_p$.

For spin one-half there is no independent bare electric stiffness within the ice manifold. In the conventional compact-gauge Hamiltonian $\frac{U}{2} \sum_\ell E_\ell^2 - K \sum_p \cos B_p$, the identification $E_\ell = S_\ell^z = \pm\frac{1}{2}$ makes $E_\ell^2 = \frac{1}{4}$ a constant. The photon stiffness is therefore generated dynamically by the ring term itself [2, 7, 22]. This is the strongly interacting limit in which we ask how the microscopic spectral weight departs from Gaussian electrodynamics.

A. The computed observable

Throughout, the quantity we compute is the zero-temperature, sublattice-summed longitudinal correlation function of the local Ising pseudospin,

$$S^{zz}(\boldsymbol{q}, \omega) = \frac{1}{N} \sum_{\mu} \sum_{n} \left| \langle n | S_\mu^z(\boldsymbol{q}) | 0 \rangle \right|^2 \delta(\omega - \omega_n), \tag{2}$$

with $S_\mu^z(\boldsymbol{q}) = \sum_{i \in \mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S_i^z$ and μ running over the four pyrochlore sublattices. Here S_i^z is the component along the local $\langle 111 \rangle$ easy axis at site i, which is the component that equals the emergent electric field, $E_i = S_i^z$. All intensities quoted below — "share of the intensity," "off-pole weight" — are fractions of $\sum_n \sum_\mu |\langle n | S_\mu^z(\boldsymbol{q}) | 0 \rangle|^2$ at the same $\boldsymbol{q}$, so they are normalized within Eq. (2) itself.

Equation (2) is not the neutron cross-section. The magnetic neutron cross-section of a non-Kramers or dipolar–octupolar spin ice involves, in addition, the ionic form factor, the g-tensor projecting the local pseudospin onto the laboratory magnetic dipole, the rotation from the four local $\langle 111 \rangle$ frames to the global frame, and the neutron polarization factor $\delta_{\alpha\beta} - \hat{q}_\alpha \hat{q}_\beta$. Those factors multiply Eq. (2) with sublattice-dependent and $\boldsymbol{q}$-dependent coefficients and are material specific; for the dipolar case they are given explicitly in Refs. [7, 26]. We do not apply them, because every statement we make is a comparison between eigenstates *at the same* $\boldsymbol{q}$ *and in the same sublattice channel*, where these prefactors are common and cancel from the ratios. They do not cancel from a comparison between different $\boldsymbol{q}$, and we therefore make no claim about the absolute momentum dependence of neutron intensity. When we write that a level "carries half the intensity" we mean half of the longitudinal pseudospin weight of Eq. (2) at that momentum. Converting to a measured cross-section for a specific material requires the material's g-tensor and is outside the scope of the pure ring model.

Our calculation is exact within the specified finite Hilbert spaces, not within the full microscopic spin model. Equation (1) conserves winding, and we work in the ground-state winding sector of the spin-ice manifold. Electric charges (spinons) are absent by construction. The magnetic sector is retained, so spectral weight outside the photon branches could in principle have magnetic rather than multiphoton character; we distinguish these possibilities below. A word on which plaquettes enter Eq. (1). A periodic cluster contains six-site closed loops of two kinds: the elementary hexagons of the infinite pyrochlore lattice, and loops that close only by winding around the torus. The latter are not plaquettes of the lattice, and including them adds ring terms with no counterpart in the thermodynamic limit; on small clusters they are the majority of all six-cycles found by a naive search. We therefore keep only loops of zero winding number, of which the 48-site cluster has exactly 48, twelve in each of the four $\langle 111 \rangle$ orientations. This is a property of the cluster geometry and is checked directly on each cluster used here; no result below depends on it being taken on faith.

III. OPERATOR-RESOLVED SPECTRUM

The probe operator has an especially direct gauge interpretation. Since $S^z = E$, the state $S^z(\boldsymbol{q})|0\rangle$ is the electric-field wave launched by the probe. In Gaussian electrodynamics this state lies entirely in the one-photon sector. In the compact microscopic model it can overlap with every odd-parity eigenstate allowed by the same quantum numbers. Resolving those overlaps, rather than only the lowest energy in each momentum sector, is the central numerical task here.

The two gauge fields also have transparent spin meanings. E_ℓ is the Ising moment on bond ℓ, whereas B_p is the flux through a hexagon. The expectation value

$\langle W_p + W_p^\dagger \rangle = 2\langle \cos B_p \rangle$ measures the coherence of the six-spin ring resonance. Since Eq. (1) contains the entire nonconstant energy in the ice manifold, changes in this coherence provide a direct energetic diagnostic of the excited states.

IV. THE ROKHSAR–KIVELSON DEFORMATION AS AN EXCITED-STATE PROBE

To classify the eigenstates we use the conventional Rokhsar–Kivelson extension of the pyrochlore ring model,

$$H(V) = -K \sum_p \left(W_p + W_p^\dagger\right) + V \sum_p n_p^{\mathrm{flip}}, \tag{3}$$

where n_p^{flip} projects onto an alternating hexagon, so that $\sum_p n_p^{\mathrm{flip}}$ simply counts the flippable plaquettes in a configuration. The sign convention matters for reading the figures and we state it once: $V > 0$ penalizes flippable plaquettes, $V < 0$ rewards them, and the exactly solvable Rokhsar–Kivelson point sits at $V = +K$, where the equal-amplitude superposition is an exact ground state. Our scans run over $V/K \in [-0.5, +0.5]$ unless stated otherwise, and the anomalous level softens toward negative V, that is, toward configurations with more flippable plaquettes. This flippability potential was introduced for the pyrochlore ring model in the original gauge-theory construction and subsequently used to map the quantum-ice phase diagram and tune the emergent electrodynamics [2, 8, 10, 27]. Our use of it is different. We remain at the physical ring point $V = 0$ and use only the infinitesimal derivative of an individual excitation energy. Hellmann–Feynman gives

$$\frac{d\omega_n}{dV} = \langle n| \sum_p n_p^{\mathrm{flip}} |n\rangle - \langle 0| \sum_p n_p^{\mathrm{flip}} |0\rangle, \tag{4}$$

so the slope is an exact state-resolved census of the number of flippable hexagons added or removed by the excitation. No parameter is fit to obtain this diagnostic.

At L the two strongest S^z-active levels respond with opposite signs. The level at $1.73K$, close to the Gaussian-photon energy $2.03K$, loses flippable hexagons as V is increased and behaves as an ordinary dressed photon. The lower level at $1.02K$ carries 51% of the S^z spectral weight and gains flippable hexagons. It is also the lowest eigenvalue in the fully diagonalized 9104-dimensional 48-site ground-state winding sector, including both spin-reversal parities. We do not elevate that finite-sector statement to a claim about all winding sectors or about the full microscopic spin model. Parity excludes only even photon number; the three-photon channel is excluded separately, and by a wide margin, on kinematic grounds (Sec. VIII), while the flux-correlation diagnostic discussed below disfavors a localized magnetic pair. The combination of collective ancestry, strong finite-momentum softening, and qualitatively different microscopic response motivates the roton-like terminology.

V. OSCILLATOR STRENGTH AND THE FEYNMAN RATIO

The first moment of the neutron structure factor can be evaluated exactly in the ring model:

$$f_1(\boldsymbol{q}) = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu \left| g_p^\mu(\boldsymbol{q}) \right|^2, \tag{5}$$

where $g_p^\mu(\boldsymbol{q})$ is the lattice curl, equivalently the flux driven through hexagon p by the electric-field wave. Equation (5) has the form of a Maxwell magnetic energy: the geometric factor is the squared curl of the wave, while the local stiffness is the measured ground-state ring coherence $\langle W_p + W_p^\dagger \rangle$. It contains no phenomenological stiffness parameter, reproduces the exact spectral first moment at every sampled momentum, and satisfies $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$ identically.

The normalized single-mode state proportional to $S^z(\boldsymbol{q})|0\rangle$ therefore has variational energy

$$\omega_{\mathrm{SMA}}(\boldsymbol{q}) = \frac{f_1(\boldsymbol{q})}{S(\boldsymbol{q})}, \tag{6}$$

with $S(\boldsymbol{q})$ the equal-time structure factor. This is the standard Feynman logic [28]: a collective mode is inexpensive where its oscillator strength is small and the corresponding static fluctuation is already large. The logic itself is not new; it is also familiar from Rokhsar–Kivelson quantum dimer models [27, 29]. What is specific here is the microscopic origin of both factors at the physical $V = 0$ ring point.

A. What the ratio does and does not establish

It is essential to be clear about the logical status of Eq. (6). The ratio f_1/S is, identically, the first moment of $S^{zz}(\boldsymbol{q}, \omega)$ divided by its zeroth moment: it is the *centroid* of the spectrum created by $S^z(\boldsymbol{q})$. Observing after the fact that the centroid is low where a heavy low-lying pole exists is therefore not evidence for anything; it is a restatement. We do not use it that way.

The content of Eq. (6) is that both of its factors are properties of the *ground state alone*. f_1 follows from the double commutator and is fixed by the ground-state plaquette coherences $\langle W_p + W_p^\dagger \rangle$ and lattice geometry; $S(\boldsymbol{q})$ is the equal-time correlator. Neither requires knowledge of any excited state. The ratio is thus usable as a predictor, evaluated before the spectrum is computed, and the meaningful tests are those in which it predicts something not already known.

Four such tests are available here, and all are reported below. The first is a class prediction: the anisotropic 64-site cluster contains two inequivalent L-type momentum

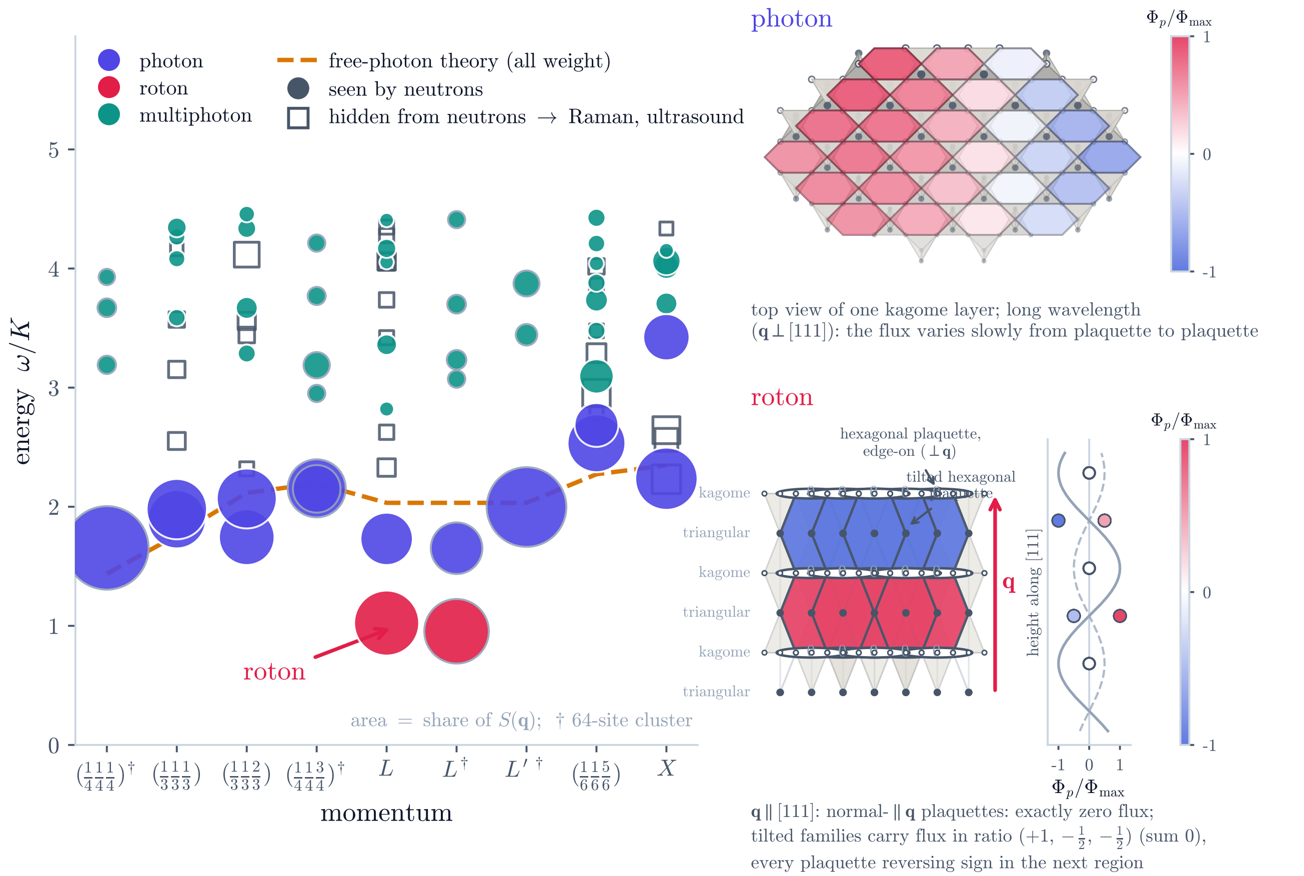

FIG. 1. Operator-resolved spectrum of the ring model in the ground-state winding sector and diagnostics of the dominant modes. Left: S^z-active levels at nine inequivalent momenta (48-site cluster; † marks momenta from the 64-site cluster). Marker area is the fraction of the $S^z(\boldsymbol{q})$ spectral weight at that momentum carried by the level. Open squares denote spin-reversal-even levels, which are exactly dark to S^z and are shown for comparison with bilinear probes. The two photon descendants are identified by projection onto the transverse Gaussian polarizations. The anomalous L-point level is distinguished by two separately computed properties: its energy is about half the Gaussian-photon value and its flippability response is positive, $\Delta\langle n^{\rm flip}\rangle = +0.34$, whereas the ordinary photon descendants do not appreciably gain flippable hexagons. At L it appears at $1.02K$ with 51% of the S^z spectral weight on 48 sites and at $L^\dagger$ at $0.95K$ with 50% on 64 sites. The inequivalent L' class of the anisotropic 64-site box remains photon-dominated when its equal-time structure factor is suppressed. The dashed curve is the Gaussian photon evaluated on the same lattice. Right: flux patterns associated with a long-wavelength photon and the shortest sampled wave along [111]. At the latter momentum, the hexagons lying in wavefronts enclose exactly zero flux; the three tilted families carry flux in the ratio $(+1, -\frac{1}{2}, -\frac{1}{2})$ and reverse sign between adjacent interlayer regions. Higher S^z-active levels carry spectral weight absent from the Gaussian one-photon description.

classes with substantially different $S(\boldsymbol{q})$, and the ground-state ratio selects, in advance, which of the two reconstructs. The second is the same prediction made independently on the 96-site cluster, whose L star also splits; it succeeds again. The third is quantitative: on the 72-site cluster two *different* momentum stars happen to have ratios within 0.9% of one another, and their soft levels come out within 0.7% — a ground-state quantity predicting two excitation energies to better than a percent. The fourth is the trajectory under the Rokhsar–Kivelson bias: $f_1(L)$ is constant to better than one percent while $S(L)$ grows and the level softens, so the ground-state factor moves first and the spectral response follows. Statements of the form "the centroid is low at L" carry no weight on their own and are not used as evidence in what follows.

VI. WHY THE SOFTENING OCCURS AT L

Two ingredients coincide at this momentum. The first is geometric. A hexagon contained in a wavefront, i.e. in a plane of constant phase perpendicular to $\boldsymbol{q}$, encloses zero flux from the electric-field wave: the alternating loop sum defining g_p vanishes exactly. Such a plaquette contributes nothing to Eq. (5). Pyrochlore hexagons are perpendicular to the four $\langle 111\rangle$ directions, and at the sampled zone-boundary point $L \parallel [111]$ one of the four hexagon families lies in the wavefronts. That family therefore drops out of the oscillator-strength cost exactly, and the vanishing is an identity, not a small number: resolved by family, the geometric factor $\sum_p \sum_\mu |g_p^\mu(\boldsymbol{q})|^2$ at L is $(96, 96, 96, 0)$, against $(96, 96, 96, 96)$ at X. The reduction of the *geometric* factor is therefore exactly 3/4. The reduction of f_1 itself is not exactly 3/4, because Eq. (5) weights each plaquette by its own ring coherence $w_p = \langle W_p + W_p^\dagger \rangle$, which is not uniform on an anisotropic cluster; numerically $f_1(L)/f_1(X) = 0.684/0.880 = 0.777$.

It is important that this geometric decoupling is necessary but not sufficient. It occurs at every momentum along [111]: at the smallest such momentum the family resolution is $(72, 0, 72, 72)$, again with one family exactly dark. Yet that momentum shows no anomaly, its dominant level sitting at $1.98K$ against a Gaussian value of $1.76K$. What distinguishes L is the second ingredient below.

The second ingredient is dynamical. In the interacting $V = 0$ ground state the equal-time structure factor is largest at L among the sampled momenta: $S(L) = 0.370$, compared with 0.19–0.29 elsewhere. The combination of reduced f_1 and enhanced S makes f_1/S smallest at L among the twelve cluster momenta and makes it the only sampled point where the single-mode energy falls below the Gaussian-photon line.

Crucially, the enhancement of $S(L)$ is not inherited from the ice constraint alone, and two independent controls establish this.

The first is exact and requires no sampling. The equal-amplitude superposition over the *same* 9104 configurations of the *same* cluster is the classical ice ensemble restricted to exactly what the quantum state can reach, so comparing it with the interacting ground state isolates the effect of the amplitudes alone. It gives $S(L) = 0.297$ against the interacting value 0.370. Measured as an enhancement over the other sampled momenta, the classical ensemble raises L by 11% and the interacting ground state by 48%. Roughly four fifths of the L enhancement is therefore made by the quantum dynamics, not by the ice rule.

The second checks that this is not an artifact of the small cluster. Loop-update Monte Carlo on a 2048-site classical ice manifold, binned into 30 independent blocks, gives $S(L) = 0.2488(67)$, $S(X) = 0.2538(73)$, $S(W) = 0.2475(52)$ and $S(K) = 0.2467(46)$. The total spread across these zone-boundary momenta, 0.007, is comparable to the typical statistical error, 0.006: within the resolution of the calculation the classical structure factor is flat, and L in particular is not distinguished. (The zone center is excluded from this comparison; it is the uniform mode, and it carries no weight in the finite cluster.) The much stronger L enhancement seen in the interacting ground state is generated by the quantum ring dynamics at $V = 0$. This is the main distinction from an RK-point resonon, whose softening can be read directly from the classical correlations of the equal-amplitude state.

Finally, the mechanism is the *ratio*, and L is where the ratio is minimized rather than a privileged point in itself. The 72-site cluster makes this concrete: its momentum $(\frac{1}{6}, \frac{1}{2}, \frac{1}{2})$ has a family decomposition $(36, 144, 144, 108)$ — one hexagon family suppressed to a quarter rather than the exact zero that L enjoys — but a structure factor almost as large, and its ratio comes out within 0.9% of the L value. Its spectrum is correspondingly almost identical: a dominant level at $1.075K$ with 55% of the weight, against $1.068K$ with 54% at L. Exact geometric decoupling is therefore one route to a small ratio; partial suppression combined with a large $S(\boldsymbol{q})$ is another, and the spectrum follows the ratio, not the route.

VII. MICROSCOPIC CHARACTER OF THE LOW L MODE

The state-resolved expectation values show that flippability and resonance coherence are different observables. Relative to the ground state, the low L level contains 0.34 more flippable hexagons, yet its total magnetic coherence $\sum_p \langle \cos B_p \rangle$ is smaller. Because the ring term is the full nonconstant Hamiltonian in the constrained manifold, the loss of coherence accounts directly for the excitation energy. In spin language, more plaquettes are instantaneously arranged so that they *could* flip, while the phases of those ring processes are less coherent. The mode therefore pays energy by disrupting resonance, not by violating the ice rule.

Two different single-mode energies appear in this problem and should not be confused. The Feynman ratio of Eq. (6) uses the sublattice-summed probe and gives $f_1(L)/S(L) = 0.684/0.370 = 1.85K$, which is the centroid of that state's spectrum. A better trial state is available at no extra cost: the probe resolves into four sublattice-labelled states $S_\mu^z(L)|0\rangle$, and the lowest eigenvalue of H within that four-dimensional space is $1.450K$. We take the latter as the bare single-mode energy, since it is the best description of the excitation that uses only the electric-field wave.

The exact low L eigenstate lies at $1.024K$, so relaxation beyond the electric-field wave recovers $0.427K$, or 29% of the bare single-mode cost. Because the ring term is the entire nonconstant Hamiltonian, this recovery decomposes exactly, with no remainder, into contributions from the four hexagon families (Appendix C). Of it, 71% comes from the single family lying in the wavefronts — the family whose plaquettes carry no flux from the imposed wave and therefore contribute nothing to the oscillator strength. They can reorganize without reducing the overlap with the probe state, which is precisely the mechanism of backflow in Feynman–Cohen theory [30]: the excitation lowers its energy by rearranging the degrees of freedom the bare wave does not constrain.

VIII. FEW-PHOTON KINEMATICS

Spin-reversal parity alone does not identify the low L level. Under charge conjugation an n-photon state carries parity $(-1)^n$, so the odd parity of the level excludes even photon number but permits $n = 3$. We therefore test the three-photon assignment kinematically. Using the Gaussian branch quantized on the same torus (Appendix A), we minimize the total energy over every partition of the cluster momentum into n photon momenta drawn from the allowed set, excluding Γ, which carries no photon because the curl has no nonzero singular value there. At L the resulting thresholds are

$$\omega_L^{(1)} = 2.03K, \qquad \omega_L^{(2)} = 4.03K, \qquad \omega_L^{(3)} = 5.55K. \quad (7)$$

The level in question lies at $1.02K$, a factor of 5.4 below the three-photon threshold and below even the one-photon energy. The same ordering holds at every sampled momentum: the three-photon threshold never falls below $5.28K$, because the softest photon available on this cluster already costs $1.76K$. A three-photon interpretation of the L level is therefore excluded on the clusters we can diagonalize.

This argument uses the Gaussian dispersion to count the cost of the constituents, and so it is an argument about weakly renormalized photons. It does not by itself exclude a multiparticle state built from strongly renormalized constituents. What makes that alternative implausible here is that the level is a single, sharp, dominant pole carrying half of the weight at its momentum rather than the small distributed weight characteristic of a continuum, and that its polarization content is that of a single transverse branch (Appendix B).

IX. IN WHAT SENSE THIS IS A ROTON

The word carries several meanings and we state exactly which one we claim. Operationally, a roton-like minimum is a finite-momentum minimum of a collective excitation branch, produced by the Feynman mechanism — an excitation is cheap where the static structure factor is large and the oscillator strength small — and relaxed further by backflow-like reorganization of the degrees of freedom the bare density wave does not constrain. Every element of that definition is now established by exact or gated computation: the lowest bright excitation of every diagonalized cluster lies at L, below the photon branch at all other accessible momenta (Sec. XII and the four-cluster table below); the ground-state ratio f_1/S predicts the location, three times, forward (Sec. V A); its zone-boundary minimum at L persists to 2048 sites; and the plaquette-resolved relaxation is dominated by the geometrically dark hexagon family (Appendix C).

Two things are *not* claimed, and the distinction is part of the result rather than a hedge. First, sharpness beyond 96 spins: exact diagonalization establishes a single dominant pole up to that size, while the Monte Carlo constrains only the centroid; whether the pole broadens into a narrow continuum in the thermodynamic limit is open, and the gapless theory permits multi-quantum weight to descend toward any branch. What is size-stable is where the spectrum is soft and how much weight is there. Second, the radial profile: on the lattice the [111] line terminates at L, so the minimum is over the zone-boundary landscape and over the bright spectrum, not a dip along a radial cut as in helium. The analogy to helium is nevertheless the correct mechanical one. There, the high-momentum continuation of the sound branch is softened where $S(q)$ peaks and lowered further by backflow. Here the electric field is microscopically saturated, $E = S^z = \pm\frac{1}{2}$, so a finite-amplitude wave cannot be formed by increasing $|E|$ on a bond: it must rearrange which bonds carry the two allowed values, that rearrangement costs ring coherence, and the cost is smallest at L — for the geometric and dynamical reasons of the preceding sections.

We add what the identification does not rest on. It does not rest on photon ancestry: with E^2 a c-number there is no interpolation to a free theory, so no adiabatic branch labeling exists, and we call the mode only the dominant excitation in one transverse S^z channel. It does not rest on the fitted Gaussian scale: the scale-free branch ratio 0.55–0.59 and the zone-boundary hierarchy of f_1/S involve no fitted quantity. And it does not rest on the flippability census, which is a consistent signature but not size-stable in magnitude (Sec. XIII).

X. RELATION TO RK RESONONS

Single-mode soft modes are familiar in quantum dimer models at Rokhsar–Kivelson points. There the ground state is an equal-amplitude superposition and its static correlations are those of the corresponding classical dimer ensemble [27]. On the square lattice this produces the well-known resonon structure, confirmed in dynamical calculations [29]. The three-dimensional cubic RK model is different: its only soft mode is the remnant of the photon, consistent with the comparatively featureless classical Coulomb-phase correlations [29]. Our L anomaly is therefore not an RK-point soft mode transplanted to pyrochlore. It occurs at the physical $V = 0$ ring point, remains at finite energy on the available clusters, and relies on an $S(L)$ enhancement generated by the interacting quantum ground state rather than by the equal-amplitude classical ensemble.

XI. FINITE-SIZE AND DEFORMATION CHECKS

Convergence of the collective state. A systematic variational improvement of the electric-field wave converges rapidly to the exact 48-site eigenvalue, $1.450 \to 1.074 \to$

$1.034 \to 1.026K$ against $1.024K$ from complete diagonalization. The state is reached from the probe in a few steps, as a collective mode should be, rather than requiring the full Hilbert space.

A prediction, not a fit. The 64-site cluster provides the one test in which the ground-state ratio is used in the direction it is supposed to work: forward. Its anisotropy splits the L star into two inequivalent classes, which the Gaussian theory cannot distinguish — both have the same $|\boldsymbol{q}|$ and the same Gaussian energy. They differ in a ground-state quantity, the equal-time structure factor: $S = 0.369$ for one class and $S = 0.227$ for the other. The f_1/S mechanism therefore predicts, before any excited state is computed, that the first class reconstructs and the second does not. It does. The class with $S = 0.369$ carries a level at $0.954K$ with 50.0% of the weight; the class with $S = 0.227$ has its dominant level at $1.998K$ with 77% of the weight, and no anomalous state at all. The effect follows the structure factor rather than the label L.

Stability with size. We can now compare four clusters. The third, 2×3×3 with 72 spins, required extending the state representation past a single 64-bit word; the fourth, 2×3×4 with 96 spins, required in addition a vectorized enumeration of its two-billion-configuration ice manifold and a Lanczos treatment of a 146 758 352-dimensional winding sector (Appendix E). Ninety-six spins matches the largest constrained-space exact diagonalization reported for this model [10], there used for ground-state properties; here the full operator-resolved response is obtained. The four boxes span three distinct shapes.

	2×2×3	2×2×4	2×3×3	2×3×4
sites	48	64	72	96
ice configurations	87 546	$2.76{\times}10^6$	$1.28{\times}10^7$	$2.01{\times}10^9$
sector dimension	9 104	249 180	973 008	$1.47{\times}10^8$
E_0/N	−0.2199	−0.2165	−0.2057	−0.2038
$S(L)$	0.370	0.369	0.321	0.355
$\omega_{\rm low}/K$	1.024	0.954	1.068	0.904
share	50.8%	50.0%	54.0%	54.4%
$\omega_{\rm high}/K$	1.730	1.652	1.814	1.647
$\omega_{\rm low}/\omega_{\rm high}$	0.592	0.578	0.589	0.549
census $d\omega_{\rm low}/dV$	+0.34	+0.34	+0.13	+0.26

The qualitative picture is identical on all four: a single dominant level near $1K$ carrying at least half of the weight at L, and a second transverse branch roughly seventy percent higher. The scale-free ratio of the two branch energies varies over 0.549–0.592 across a factor of 16 000 in sector dimension and three box shapes; the soft level's share of the weight varies over 50–54%. This is the evidence least contaminated by fitted quantities. On the largest cluster the mode is, if anything, the softest of the four.

The class prediction repeats at 96 sites. The 2×3×4 box, like the 64-site one, carries two inequivalent L classes, with $S = 0.355$ and $S = 0.241$. Computed before any excited state, the ratio again selects the first: it develops the $0.904K$ level with 54.4% of the weight, while the second remains photon-like, its two dominant levels at $1.651K$ and $1.730K$ with flippability censuses of −0.02 and −0.09 against +0.26 for the soft mode. The forward discrimination between L classes has therefore succeeded at two sizes independently, and the census contrast within a single cluster — positive for the class that reconstructs, negative at the same $|\boldsymbol{q}|$ for the class that does not — appears at both.

Not everything is stable, and we say which. $S(L)$ moves between 0.321 and 0.370 with box shape; the weight retained by the upper branch varies between 18% and 32%; the census magnitude varies by a factor 2.6 (Sec. XIII); and the polarization splitting at X is not converged, falling from 42% at 48 sites to 5.9% at 64; no conclusion is drawn from it.

The deformation trajectory. Under a bias toward more flippable configurations ($V < 0$), $f_1(L)$ changes by less than one percent while $S(L)$ grows and the level softens from $1.02K$ to $0.73K$. The ground-state factor moves first and the spectrum follows, which is the ordering the f_1/S mechanism requires and the opposite of what a fitted correlation would produce.

XII. THERMODYNAMIC EVIDENCE: THE RATIO AT 2048 SITES

Because both factors of Eq. (6) are ground-state quantities, they are computable far beyond any exactly diagonalizable size. The ring Hamiltonian is stoquastic — every off-diagonal element equals $-K$ and the diagonal vanishes — so its ground state has no sign problem, and Green's-function Monte Carlo with the equal-amplitude trial state and forward walking yields unbiased pure-state expectation values of diagonal observables (Appendix D). $S(\boldsymbol{q})$ is diagonal; and on cubic $n{\times}n{\times}n$ supercells every hexagon is symmetry-equivalent, so the exact sum rule $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$ fixes the plaquette coherence from the energy alone and $f_1(\boldsymbol{q})$ follows with no additional statistical error. The method is gated, not assumed: on the 48-site cluster it reproduces the exact E_0 to 0.5σ and $S(L) = 0.370365$ to every printed digit — decisively away from the equal-amplitude value 0.297, which is where a broken forward walker would land.

We computed $S(\boldsymbol{q})$ on the *complete* allowed momentum grids of the 6^3 (864-site, 216 momenta) and 8^3 (2048-site, 512 momenta) supercells, where every value is a true eigenmode weight with no interpolation. Averaged over symmetry copies (which scatter by less than 0.04, an internal consistency check), the Feynman ratio at the zone-boundary points is

	864 sites	2048 sites
L	2.142(10)	2.179(15)
X	2.660(10)	2.713(21)
W	—	2.781(10)
K	—	2.718(16)
U	—	2.736(8)

in units of K. The minimum of the ratio over the zone boundary sits at L at both sizes, 20–25% below every other zone-boundary point, and moves by 1.7% between 864 and 2048 sites. The large-system data were genuinely needed: interior momenta drift substantially with size — at $(\frac{1}{3},\frac{1}{3},\frac{1}{3})$ the ratio falls from 2.42 on the 48-site cluster to 1.81 at 864 sites — whereas the L and X values barely move, so the zone-boundary hierarchy could not have been extrapolated from exact diagonalization alone.

Two aspects of the geometry must be stated precisely, because a looser reading would overclaim. First, along the radial [111] direction the ratio rises monotonically from zero at Γ — 0.60, 1.36, 1.97, 2.18 at the allowed momenta $\frac{k}{8}(1,1,1)$ of the 8^3 cell — and L is the endpoint of that line: beyond it the band folds back, because $2\pi(1,1,1)$ is a reciprocal lattice vector. The minimum is therefore not a helium-style dip along a radial cut; it is a minimum of the ratio over the zone-boundary landscape, the surface on which all the competing short-wavelength excitations live, where the descent into L is steep (from 2.78 at W). Second, the ratio is the spectral centroid, an upper bound on the lowest bright excitation: the Monte Carlo therefore constrains where the spectrum is soft, and the exact diagonalization is what shows that a single sharp pole, at roughly half the centroid, is what makes it soft. The two calculations are complementary, and we do not claim either one alone establishes the mode.

A. The soft region is a valley, and the mode is not only at L

A quasiparticle minimum should be concave up, and one should ask in which directions it is, and whether the softness is confined to a single point. The complete grid answers both. Away from L the ratio rises in *every* allowed non-radial direction: to 2.216(5) at $(\frac{1}{4},\frac{1}{2},\frac{1}{2})$, to 2.265(5) at $(\frac{3}{8},\frac{3}{8},\frac{5}{8})$, to 2.484(5) at $(\frac{1}{4},\frac{1}{2},\frac{3}{4})$, and steeply to 2.78 at W — so L is a genuine minimum of the landscape, not a saddle among the zone-boundary momenta. (Along the radial [111] line the point is stationary by band periodicity, as at any zone corner; a lattice minimum lives in the boundary manifold, exactly as the magnon roton-like minima of frustrated antiferromagnets do at their zone corners.)

The rise is, however, strongly anisotropic: the landscape at 2048 sites has a shallow *valley* along the $(t,\frac{1}{2},\frac{1}{2})$ lines, with the ratio varying by only two percent from $(0,\frac{1}{2},\frac{1}{2})$ through $(\frac{1}{4},\frac{1}{2},\frac{1}{2})$ to its minimum at L, while every direction transverse to the valley climbs by 15–28%. The softness is therefore not an isolated point: it is the bottom of a flat trough. And this is not only a statement about the centroid: the 72-site cluster carries a momentum on exactly this line, $(\frac{1}{6},\frac{1}{2},\frac{1}{2})$, and its *exact* spectrum shows a soft level at $1.075K$ with 55% of the weight — essentially degenerate with the L mode at $1.068K$ (Sec. V A). The prediction that the trough hosts soft modes away from L is thus verified spectrally at the one accessible momentum on it. A flat valley of nearly degenerate soft collective modes, rather than a single deep pocket, is itself a statement about the physics: whatever ordering tendency the softening signals is frustrated among the wavevectors of the valley.

XIII. WHAT THE FLIPPABILITY CENSUS DOES AND DOES NOT SHOW

It is tempting to read the census as a sign rule: an excitation disrupts ring resonance, so it should *lose* flippable plaquettes, and a level that gains them is thereby marked as different in kind. That reading is too strong, and the exact statement is worth having because it says precisely how much is true.

The operator $N=\sum_p n_p^{\mathrm{flip}}$ is diagonal in the configuration basis, and each flippable hexagon of a configuration contributes exactly one off-diagonal element of the ring Hamiltonian. Hence $\mathrm{Tr}\,N$ equals the number of nonzero off-diagonal elements of H, and the census averaged over *all* eigenstates of a sector of dimension D is

$$\frac{1}{D}\sum_n \frac{d\omega_n}{dV} = \frac{\mathrm{nnz}(H)}{D} - \langle 0|N|0\rangle, \qquad (8)$$

which is negative precisely because the ground state is more flippable than a typical ice configuration. On the 48-site cluster Eq. (8) gives -0.9139, and direct diagonalization of all 9104 eigenstates returns the same number to 10^{-14}.

So excitations do lose flippability on average, exactly and provably. Nothing, however, constrains an individual level. Of the 9104 eigenstates, 58 — 0.64% — have a positive census, and 20 exceed $+0.20$. The anomalous L level at $+0.339$ sits in that top 0.22% tail, which is the honest form of the statement: it is extreme, not unique.

The census is also the least size-stable of our diagnostics, and we say so plainly. At the anomalous level it is $+0.339$ on 48 sites, $+0.338$ on 64, $+0.134$ on 72 and $+0.259$ on 96. The sign is robust; the magnitude is not, and no normalization we tried repairs it. On 48 and 64 sites the anomalous level is the only appreciably positive bright census, exceeding the next by factors of 29 and 7; on 72 sites several bright levels are positive and it is no longer the largest. We therefore do not use the census as the primary discriminant.

What does survive on every cluster is a comparison made *within* one of them, and it now appears at two sizes. The 64- and 96-site boxes each carry two inequivalent

L-type classes at identical $|\boldsymbol{q}|$ and identical Gaussian energy. The class that reconstructs has census $+0.338$ (64) and $+0.259$ (96); the class that does not has -0.166 and $-0.02/-0.09$ respectively. Same momentum star, same free-theory prediction, opposite response to the flippability probe. That contrast needs no comparison across cluster sizes and no fitted scale, and it is the census evidence we rely on.

XIV. POLARIZATION SPLITTING AND OFF-POLE WEIGHT

The two transverse polarizations are exactly degenerate in the Gaussian theory, and their interaction-induced splitting was predicted in semiclassical treatments [22, 23]. Our calculation determines its magnitude in the microscopic ring model without an expansion. Because the two brightest levels at each momentum are polarization-pure and mutually orthogonal (Appendix B), the splitting can be read off directly as the energy difference between them. It grows monotonically as the momentum increases: it is unresolved, below the $10^{-8}K$ level-grouping tolerance, at the smallest momentum of the 64-site cluster, 4.4% at the smallest momentum of the 48-site cluster, 17% at $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$, 42% at X and 51% at L. The degeneracy is thus restored in the infrared, as the Gaussian theory requires, and destroyed at the zone boundary.

We emphasize what this does *not* add. The splitting at L and the roton-like softening at L are the same fact: the softened level is the lower of the two transverse branches, and the splitting is the size of the softening. It is reported here because it makes contact with a specific earlier prediction, not as independent evidence for the anomaly.

Beyond the two photon descendants, 16–38% of the S^z spectral weight resides in higher levels absent from the Gaussian one-photon theory. This fraction is defined without any energy window or fitted scale: the two photon descendants are identified as the brightest level in each transverse polarization channel, and the off-pole weight is everything else (Appendix B). An alternative operational definition — simply the two brightest levels at that momentum — gives identical numbers at every momentum sampled. The off-pole fraction takes its smallest value, 16% on 48 sites and 13% on 64 sites, at the smallest momentum available on each, consistent with recovery of the infrared photon.

Several diagnostics favor a multiparticle rather than a localized magnetic interpretation of this higher weight. Plaquette-flux correlations are weak and spatially extended instead of showing the localized disturbance expected for a bound magnetic pair, and the response to the RK deformation is consistent with sums of single-mode responses. By contrast, leading-order perturbation theory in the quartic term obtained by expanding the cosine predicts polarization splittings and transferred spectral weight below 0.1% at the coupling generated by this model, and only a few percent even when that coupling is artificially tripled. The finite-momentum spectrum is therefore not quantitatively described as a weak anharmonic correction to the Gaussian photon.

XV. PARITY AND COMPLEMENTARY PROBES

Global spin reversal leaves Eq. (1) invariant. Eigenstates can therefore be chosen even or odd under this symmetry, while S^z is odd. Consequently a spin-reversal-even ground state couples through neutron scattering only to odd states. This elementary selection rule is exact in the model; numerically every even level carries S^z weight below 10^{-28}. In gauge language it is charge conjugation, under which an n-photon state has parity $(-1)^n$.

The practical implication is a clean division between linear and bilinear probes. Even-sector weight, including the two-photon channel, is invisible to S^z but can appear in probes coupling to spin pairs, such as Raman scattering [31] and ultrasound [32]. Stray-field noise magnetometry [33], which couples to magnetization, belongs to the neutron-like odd channel. The selection rule itself is not a new symmetry statement; its value here is that the complete finite-size spectrum makes the complementary experimental channels explicit.

XVI. EXPERIMENTAL IMPLICATIONS

The calculation does not imply that every zone-boundary feature in a microscopic quantum-spin-ice material is the same as the low L level of the pure ring model. Longer ring exchanges, virtual spinons, and material-specific interactions can modify the spectrum. It does show, however, that even the minimal compact gauge Hamiltonian can redistribute a large fraction of the S^z spectral weight away from the Gaussian photon without leaving the deconfined phase. At the sampled L point the dominant level lies at roughly half the Gaussian-photon energy and carries about half of the weight, while the other transverse branch retains only a minority. More generally, the zone-boundary response contains split photon branches and substantial off-pole intensity.

Stated as something to look for, the pattern is this. A spectrum that is a single photon pole near the zone center need not remain a single pole at the zone boundary: along [111] the weight can move onto a lower branch that sits well below the linear extrapolation of the small-$\boldsymbol{q}$ dispersion, while a weaker feature persists near where the extrapolation predicts. The two should appear at the same momentum with comparable widths and a weight ratio of roughly two to one, and the lower one should be the more intense. Because the splitting is between the two transverse polarizations, polarized neutron scattering separates them directly, which is a sharper test than the unpolarized lineshape. We caution that a finite cluster produces discrete poles where the thermodynamic system may produce a peak plus a continuum, so the robust prediction is the redistribution of weight, not the pole count.

This provides a microscopic candidate for the broad finite-momentum response seen in quantum Monte Carlo [25], but we do not identify the QMC linewidth uniquely with the finite-size levels found here. In the thermodynamic limit the gapless theory also permits multi-quantum continua to extend down toward the one-photon branch, so experimental comparison should emphasize the redistribution and lineshape of spectral weight rather than a one-to-one count of finite cluster poles.

XVII. LIMITATIONS AND CLUSTERS

Our conclusions are subject to four explicit restrictions. First, the calculation is performed in the ground-state winding sector of the spin-ice ring model, not over all topological sectors and not in the full microscopic spin Hilbert space. Second, exact spectra exist at four sizes and no more; the scale-free diagnostics — the branch ratio and the soft level's weight share — are stable across all four, but $S(L)$, the upper branch's weight and the census magnitude move with box shape, and the splitting at X is unconverged. Third, the sharpness of the mode is established only up to 96 spins. The Monte Carlo extends the ground-state predictor to 2048 sites but constrains the spectral centroid, not the poles; whether the level broadens into a narrow continuum in the thermodynamic limit is open. Fourth, longer ring exchanges and virtual-spinon processes in microscopic XXZ models lie outside Eq. (1). Whether the anomaly survives their addition is a question this calculation cannot answer, and no claim here rests on it; work on the XXZ model is in progress and will be reported separately. The overall ring scale K can be constrained thermodynamically [34].

The cluster choice itself is important. The standard 16- and 32-site pyrochlore cells are pathological for this question. On the 32-site cell every configuration in the connected zero-flux sector has exactly eight flippable hexagons, making the equal-amplitude state an exact eigenstate and producing an artificially integer-valued spectrum; on the 16-site cell only six transport sectors have any dynamics. We therefore use the $2{\times}2{\times}3$ cluster (48 sites, 87 546 ice configurations; ground-state winding sector of dimension 9104), which we diagonalize completely so that the operator-resolved weights exactly saturate the equal-time sum rule. For the size checks we use $2{\times}2{\times}4$ (64 sites, 2 756 394 ice configurations; sector dimension 249 180), $2{\times}3{\times}3$ (72 sites, 12 846 186 configurations; sector 973 008) and $2{\times}3{\times}4$ (96 sites, 2 007 260 562 configurations; sector 146 758 352), all by Lanczos and all only after reproducing every 48-site benchmark. Momentum labels are certified by exact translation projectors.

It is natural to ask why we do not simply vary the *shape* of the torus at fixed spin number, which would be the cleanest control of all. The reason is that at these sizes there is no second admissible shape. A cluster is admissible only if it carries the full complement of contractible hexagons, one per site, as the infinite lattice does. The $2{\times}2{\times}3$ and $2{\times}3{\times}3$ tori do: they have 48 hexagons on 48 sites and 72 on 72. Every other tiling of 48 or 64 sites that contains an L-type momentum is one unit cell thick in some direction — $1{\times}3{\times}4$, $1{\times}2{\times}6$, $3{\times}4{\times}1$ and their 64-site analogues — and in those the majority of six-cycles close only by winding around the torus. The $1{\times}3{\times}4$ cluster, for instance, retains just 24 contractible hexagons on 48 sites. Keeping only the contractible ones leaves a Hamiltonian with half the ring terms per site, which is a different model rather than the same model in a different box. Shape variation at fixed size is therefore not an available test here, and the finite-size evidence rests on the sequence of sizes instead.

XVIII. CONCLUSION

The pure ring-exchange model of quantum spin ice, the minimal compact gauge theory of the ice manifold, has its lowest collective excitation not at small momentum but at the zone-boundary L point: a single level at 0.90–1.07K carrying at least half of the longitudinal weight at its momentum, below the lowest bright level at every other accessible momentum, on every cluster from 48 to 96 spins. The mechanism is Feynman's. The ground-state ratio f_1/S predicts where the spectrum reconstructs — forward, four times, once to better than a percent — and sign-free Monte Carlo shows that its zone-boundary minimum at L survives, essentially unchanged, to 2048 sites. The relaxation that carries the mode below the single-mode bound is dominated, in an exact plaquette decomposition, by the hexagon family the probe leaves unconstrained: backflow, in lattice form.

We have been precise about the two statements the data do not make: sharpness beyond 96 spins, where only the centroid is constrained, and a radial helium-style dip, which the lattice geometry does not define at L. Neither weakens what is established — a roton-like minimum in the operational sense, produced by the Feynman mechanism, at the zone boundary of the emergent-photon spectrum. The same spectra quantify photon interactions in a regime where they are nonperturbative: an order-one splitting of the transverse channels at the zone boundary — the same fact as the reconstruction, seen as an energy difference — and 16–38% of the longitudinal weight outside the two dominant poles. The principal message for experiment is that finite-momentum spectroscopy of a compact spin-one-half gauge field need not resemble a single renormalized photon pole, and that its softest feature should be sought at L.

Appendix A: The Gaussian benchmark

Every statement of the form "about half the Gaussian-photon energy" refers to the following construction, which is quantized on the same finite torus as the exact diagonalization so that no continuum extrapolation enters the comparison.

Write $B_p = \sum_\ell C_{p\ell} A_\ell$, where C is the alternating lattice curl: $C_{p\ell} = \pm 1$ with alternating sign around the six links of hexagon p, and zero otherwise. Expanding the compact magnetic energy of Eq. (1) to quadratic order and adding the electric term gives

$$H_G = \frac{U}{2} \sum_\ell E_\ell^2 + K \sum_p B_p^2, \qquad [A_\ell, E_{\ell'}] = i\delta_{\ell\ell'}. \quad \text{(A1)}$$

Restricting to momentum $\boldsymbol{q}$ and using the four-dimensional sublattice Bloch basis, the normal-mode frequencies are set by the singular values $\sigma_\lambda(\boldsymbol{q})$ of the curl in that block,

$$\omega_\lambda(\boldsymbol{q}) = \sqrt{2UK}\, \sigma_\lambda(\boldsymbol{q}). \quad \text{(A2)}$$

The construction is checked, not assumed. (i) Gauge invariance: $C\, d_0 = 0$ to machine precision, where d_0 is the tetrahedron incidence matrix signed by the up/down bipartition; the largest element of Cd_0 is 0 exactly. If this fails the sign convention is wrong and nothing downstream is meaningful. (ii) At every $\boldsymbol{q} \neq \Gamma$ the block has exactly two nonzero singular values, the two transverse photon polarizations, and at Γ it has none. (iii) The Bloch bases are orthonormal, and the probe at $+\boldsymbol{q}$ pairs with modes at $-\boldsymbol{q}$, verified by projection completeness.

A single number is fitted: the overall energy scale $\sqrt{2UK}$. In the pure ring model U is generated dynamically rather than being a bare coupling, so its value is not available from Eq. (1); this is the one place where we import a scale from the data. It is determined by one linear least-squares fit of the dominant exact line against the dominant Gaussian branch across all momenta, giving $\sqrt{2UK} = 0.5868\,K$. Everything else — the *shape* of the dispersion across momenta, the degeneracy or splitting of the two polarizations at each $\boldsymbol{q}$, and the distribution of one-photon weight over momenta and sublattice channels — is parameter-free geometry. The resulting values are those plotted as the dashed Gaussian curve in Fig. 1 and used as the denominator in Fig. 2(a); at L, $\omega_G = 2.03K$.

We stress the limitation this creates. Because one scale is fitted to the exact spectrum, the Gaussian curve cannot be used to test whether the overall photon velocity is renormalized: by construction it is not. It can only be used for statements that are invariant under a uniform rescaling of the whole branch — which momenta are anomalous *relative to the others*, and whether the two polarizations split. Those are the only statements we draw from it.

Appendix B: Polarization resolution of the bright levels

At each momentum the electric field lives in the four-dimensional sublattice space, in which the lattice curl has exactly two nonzero singular values. The neutron probe is therefore confined to a two-dimensional transverse subspace; the weight found in the remaining two directions is the longitudinal residue, which must vanish because $\nabla \cdot \boldsymbol{E} = 0$ in the ice manifold. It does, at the level of 10^{-31} relative to $S(\boldsymbol{q})$ at every momentum. This is an exact gate on the decomposition, not a fit.

A caution is necessary before any polarization statement is made. The two Gaussian transverse branches are *exactly* degenerate at every momentum, so the singular-value decomposition determines the two polarization directions only up to an arbitrary rotation within that subspace. A statement such as "this level is 68% polarization 1" is consequently meaningless in isolation. The invariant content is the geometry of the levels' polarization vectors relative to one another. For a level n of multiplicity m we form the 2×2 polarization density matrix $M_n = A_n^\dagger A_n$, where A_n is the $m \times 2$ matrix of transverse amplitudes $\langle n, \alpha | S^z(\boldsymbol{q}) | 0\rangle$. Both the purity $\mathrm{tr}\, M_n^2 / (\mathrm{tr}\, M_n)^2$ and the normalized overlap $\mathrm{tr}(M_A M_B)/(\mathrm{tr}\, M_A\, \mathrm{tr}\, M_B)$ between two levels are invariant under rotations of the degenerate subspace.

The result is unambiguous. Every bright level, including multiplets of multiplicity up to four, has polarization purity 1.000000: the whole multiplet couples through a single polarization direction. And the two brightest levels at each momentum are mutually orthogonal in polarization, so a basis exists in which each is pure. They are therefore the two transverse photon descendants, identified without reference to any energy scale.

$\boldsymbol{q}$	ω_1/K	ω_2/K	splitting	overlap
$(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$	1.895	1.980	0.044	2×10^{-28}
$(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$	1.744	2.069	0.171	1×10^{-29}
$(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$	2.533	2.687	0.059	4×10^{-2}
L	1.024	1.730	0.513	3×10^{-3}
X	2.235	3.423	0.420	1×10^{-31}

The splitting column is $|\omega_1 - \omega_2| / \frac{1}{2}(\omega_1 + \omega_2)$, the convention used throughout this paper; normalizing instead by the lower branch would give larger numbers for the same data. Orthogonality is exact to machine precision at $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{1}{3}, \frac{2}{3})$ and X, where the little group forbids mixing, and holds to 0.3% at L and 4% at $(\frac{1}{6}, \frac{1}{6}, \frac{5}{6})$, where it does not. The residual mixing is small enough that the branch assignment is never in doubt.

Two consequences matter for the interpretation. First, the identification of the anomalous L level as a photon descendant, and not a new species, rests on this decomposition and is independent of the fitted Gaussian scale. Second — and this must be stated plainly — the "interaction-induced polarization splitting" and the "roton-like softening" at L are not two independent findings. They are the same fact viewed twice: the lower of the two transverse branches at L is the anomalous level, and the splitting between the branches is the amount by which it has been pushed down. We report the splitting because it connects to a specific earlier prediction [22, 23], not as separate evidence.

Appendix C: Exact decomposition of the backflow energy

Because the ring term is the entire nonconstant Hamiltonian inside the ice manifold, the excitation energy of any state $|n\rangle$ is exactly a sum of plaquette contributions,

$$\omega_n = -K \sum_p \Big[\langle n|W_p + W_p^\dagger|n\rangle - \langle 0|W_p + W_p^\dagger|0\rangle\Big], \quad \text{(C1)}$$

with no remainder. Grouping the plaquettes by hexagon family — the four $\langle 111\rangle$ orientations, labeled unambiguously by the sublattice each hexagon omits, which is immune to the periodic wrapping that corrupts plane fitting on a small torus — turns Eq. (C1) into a four-term decomposition of any energy. Applying it to the bare single-mode state — the lowest eigenvector of H within the four-dimensional space spanned by the sublattice-resolved probe states $S^z_\mu(L)|0\rangle$, whose energy is $1.450K$ — and to the exact lowest L eigenstate at $1.024K$ gives, in units of K,

family	1	2	3	4	total
single mode	0.422	−0.540	1.004	0.565	1.450
exact	0.341	−0.476	0.702	0.456	1.024
relief	0.080	−0.064	0.301	0.110	0.427

where family 3 is the one with $g_p(L) = 0$ identically. The totals reproduce the variational and exact energies, so the decomposition is complete rather than approximate. Of the $0.427K$ of energy recovered in going from the single-mode state to the exact eigenstate, $0.301K$, or 71%, comes from the dark family alone — although it constitutes one quarter of the plaquettes and contributes nothing to the oscillator strength. This is the precise sense in which the relaxation is backflow-like: the state reorganizes exactly those degrees of freedom that the bare electric-field wave does not constrain, and which therefore cost it no overlap.

Appendix D: Green's-function Monte Carlo

The ring Hamiltonian restricted to the ice manifold is stoquastic: its diagonal vanishes and every nonzero off-diagonal element equals $-K$. The power method with the propagator $\Lambda - H$, Λ larger than the top of the spectrum, therefore converges to the ground state with strictly nonnegative amplitudes, and importance sampling with the equal-amplitude trial state $|\Psi_T\rangle = \sum_x |x\rangle$ gives walkers whose weights never change sign. A walker at configuration x moves to one of its $n^{\text{flip}}(x)$ ring-connected neighbors or stays, with the local weight $\Lambda-(\text{diagonal}) = \Lambda$ and off-diagonal flow $K\,n^{\text{flip}}(x)$; population control uses comb reconfiguration at fixed walker number, and the mixed estimator gives $E_0 = -K\langle n^{\text{flip}}\rangle_{\text{mix}}$.

$S(\boldsymbol{q})$ is diagonal in the configuration basis, and for diagonal observables the mixed distribution $\Psi_0\Psi_T$ is converted to the pure one $|\Psi_0|^2$ by forward walking: an observable measured at time t is weighted by the number of descendants its walker retains at $t+T$. We use $T = 80$ steps, beyond which the estimates are flat.

Initialization has one subtlety worth recording. Every sublattice-uniform ice state is a frozen point of the dynamics — each hexagon meets its three sublattices twice with opposite alternation parity, so no hexagon is flippable — and it is also maximally polarized, so its winding sector contains no other state and sector-preserving loop updates cannot leave it. The seed is therefore randomized by worm-constructed loop flips in two phases: first accepting loops that monotonically reduce the integer polarization $\sum_i (2n_i - 1)\,4\boldsymbol{r}_i$ (with a brief relaxation of monotonicity when stalled), then, once the minimal reachable polarization is attained, loops that preserve it exactly. Not every box reaches zero: on fcc$(2,3,3)$ the zero-polarization sector is empty, a fact known exactly from the 72-site diagonalization, and the energy gate below is what certifies the sector reached.

Every production number is gated against exact diagonalization. On the 48-site cluster the method reproduces $E_0 = -10.556012$ to 0.5σ and $S(L) = 0.370365$ to every printed digit; the equal-amplitude value of $S(L)$ is 0.297, so a defective forward walker could not pass. On the 72-site cluster it reproduces $E_0/N = -0.205690$ to 1.8σ in the correct nonzero-polarization ground sector. Statistical errors are from binned averages (16 bins); f_1 carries no statistical error beyond that of E_0, since on cubic supercells the sum rule fixes the uniform plaquette coherence as $w_p = -E_0/(K\,n_{\text{hex}})$ and the remaining factor is exact geometry. Production runs used 1024 walkers and 30 000–60 000 steps; the 2048-site full-grid run measures $S(\boldsymbol{q})$ at all 512 allowed momenta as a single matrix product per snapshot.

Appendix E: Numerical methods

Ice configurations are enumerated exactly by depth-first assignment, one site at a time, backtracking as soon as any tetrahedron would acquire a third spin of either sign, and are stored as bit masks. The ring Hamiltonian is built sparsely: for each hexagon a bit mask and the two flippable bit patterns identify the acted-on states in one vectorized pass, and the images are located by binary search in the sorted state list. The winding sector is identified from the polarization $\sum_i (n_i - \frac{1}{2})\boldsymbol{r}_i$, which is conserved by Eq. (1); we retain the sector carrying the ground state.

On the 48-site cluster the ground-state winding sector has dimension 9104 and is diagonalized completely with a dense symmetric eigensolver, so all 9104 eigenpairs are available and the operator-resolved weights saturate the equal-time sum rule $\sum_n |\langle n|S^z_\mu(\boldsymbol{q})|0\rangle|^2 = S_\mu(\boldsymbol{q})$ exactly rather than to a tolerance. On the 64-site cluster the sector has dimension 249 180 and is treated by Lanczos with full reorthogonalization, the dynamical response being obtained by a continued-fraction expansion seeded with $S^z_\mu(\boldsymbol{q})|0\rangle$; all 48-site benchmarks are reproduced before any 64-site number is quoted.

Going beyond 64 spins requires changing the state representation. Storing a configuration in a single 64-bit word caps the calculation at 64 sites, which is where the published constrained-space spectroscopy has stood. We therefore store each configuration as a *pair* of 64-bit words and redefine masking, flipping, sorting and lookup on that pair. Lookup is the only part that is not immediate: the table is sorted by the pair lexicographically, and because the high word takes fewer than 2^8 distinct values at these sizes, queries are grouped by high word and each group resolved with a single vectorized binary search on the low word within its block. The result is exact, not hashed. The implementation is validated by running it on the 48-site cluster, where it reproduces the 64-bit pipeline exactly: $E_0 = -10.55601176K$, $S(L) = 0.37036$ and branch energies $1.0235K$ and $1.7303K$.

On the 72-site cluster the ice manifold contains 12 846 186 configurations in 310 winding sectors. The ground state lies in the zero-polarization sector, of dimension 973 008; we verify this rather than assume it, by diagonalizing the largest sectors independently and comparing, which also exposes the expected exact degeneracy between symmetry-related sectors ($E_0 = -14.80971536K$ in two of them). The dynamical response is again obtained by continued fraction seeded with the four sublattice components of the probe. Because those four seeds produce separate lines at the same energy, they are summed by energy before any branch energy or weight is quoted; only the channel sum is basis-independent.

The 96-site cluster required two further steps. The two-billion-state ice manifold is enumerated by a vectorized frontier that imposes one tetrahedron at a time, visiting at each step the tetrahedron that introduces the fewest new sites and branching the frontier into independent seeds so that peak memory is bounded by one branch (a hundredth of a gigabyte) rather than the whole frontier; and exact state lookup is a single lexicographic `searchsorted` on the (hi, lo) word pairs, since at this size a block-by-block search over the ∼millions of distinct high words is no longer viable. The winding sector is identified from an exact integer polarization key without sorting the manifold, candidates estimated from a four-million-state sample and each extracted in one masked pass; the ground sector (dimension 146 758 352, 2.7×10^9 nonzeros) is identified by diagonalizing the largest candidates and comparing. The continued fraction uses real Lanczos on the real and imaginary parts of each probe separately, which is exact for a real symmetric H and halves the Krylov storage; 120 iterations reproduce the 250-iteration complex results on the 72-site cluster to every digit.

Momentum labels are certified by explicit translation projectors rather than inferred from position in a sorted list. Degenerate levels are grouped with a tolerance of $10^{-8}K$ and their weights summed, since only the total weight of a degenerate multiplet is basis-independent.

Three gates are applied throughout and reported with the data: the plaquette sum rule $\sum_p \langle W_p + W_p^\dagger \rangle = -E_0/K$, satisfied to 10^{-6}; the exactness of Eq. (5) against the directly computed spectral first moment, ratio 1.0000; and the gauge identity $C\, d_0 = 0$ of Appendix A. The scripts that produce every number and figure in this paper are included with the submission.

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We thank the Digital Research Alliance of Canada for computational resources. The complete set of scripts that generates every number, gate and figure reported here is included with the submission.

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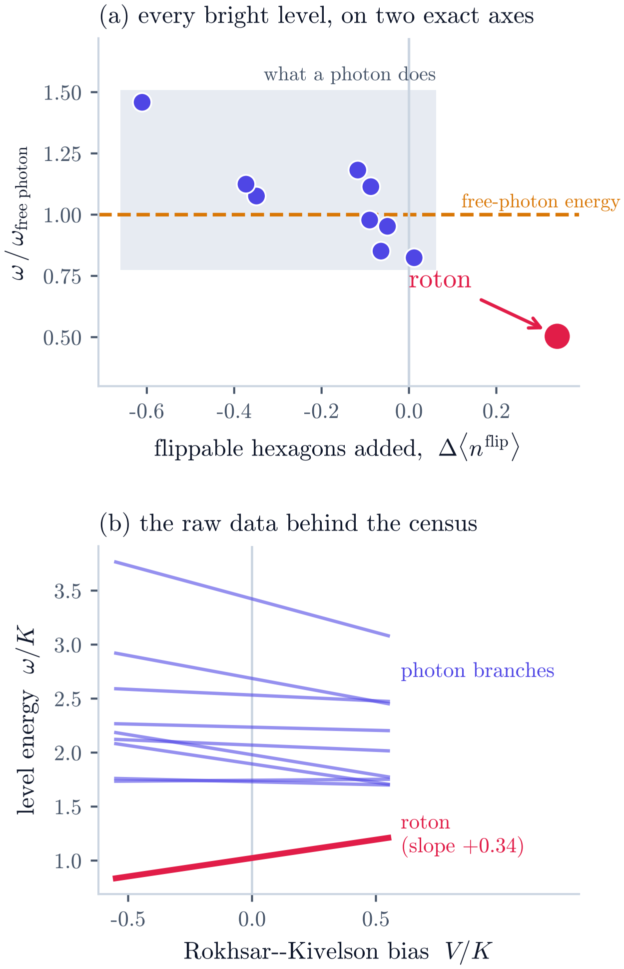

FIG. 2. State-resolved diagnostic of the anomalous L-point mode. (a) Every S^z-active 48-site level is placed on two separately computed axes. The vertical coordinate is its energy divided by the Gaussian-photon energy at the same momentum, which involves one fitted overall scale and no other parameter (Appendix A). The horizontal coordinate is the Hellmann–Feynman flippability census of Eq. (4). The photon descendants lie within 0.82–1.46 of the Gaussian energy, and on this cluster only one other bright level carries a positive census at all, +0.012, a factor of 29 below the anomalous one. The L-point roton-like level is anomalous on both axes: approximately one-half of the Gaussian energy and $\Delta\langle n^{\text{flip}}\rangle = +0.339$. (b) Evolution under the standard RK bias V. The plotted census is evaluated exactly at $V = 0$; finite-difference slopes agree to about 0.01 away from level crossings. We stress that the sparseness of positive censuses is specific to this cluster: at 72 sites several bright levels carry a positive census and the anomalous one is no longer the largest (Sec. XIII).

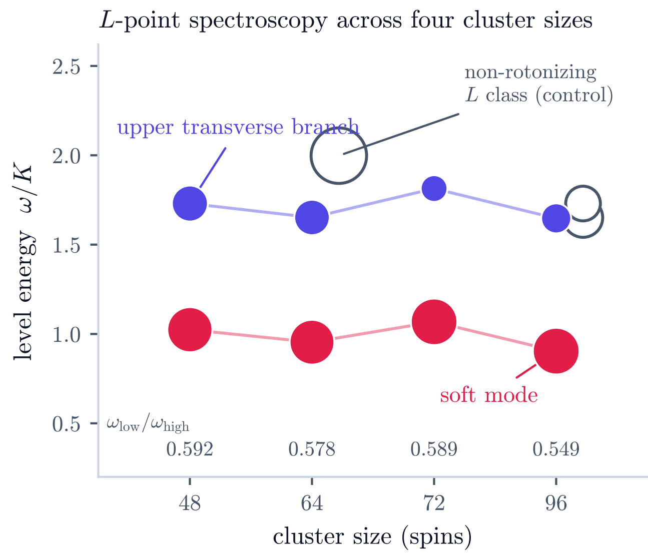

FIG. 3. The L-point spectrum across all four diagonalized cluster sizes. Marker area is the level's share of the longitudinal weight at its momentum. Filled: the soft mode (red) and the upper transverse branch (indigo) of the rotonizing L class. Open: the dominant levels of the *non*-rotonizing L class on the two clusters that possess one (64 and 96 sites) — the within-cluster control at identical $|\boldsymbol{q}|$ and identical free-theory energy. The numbers along the bottom give the scale-free branch ratio $\omega_{\rm low}/\omega_{\rm high}$. All values are exact diagonalization; the Lanczos runs carry an exact sum-rule gate that reads 1.0000 at every size (Appendix E).

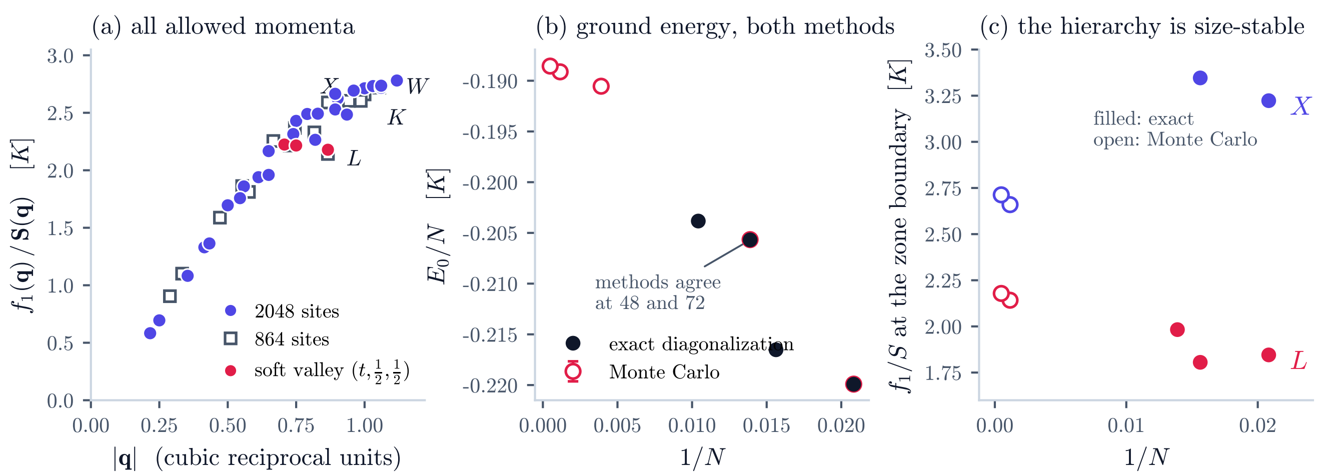

FIG. 4. Thermodynamic evidence, with the numerical checks displayed rather than asserted. (a) The Feynman ratio f_1/S at *every* allowed momentum star of the 2048-site supercell (filled circles, with statistical error bars from binned Monte Carlo averages; symmetry copies of each star agree to better than 0.04 and are merged), overlaid with the 864-site values (open squares). No path or interpolation is involved: each point is a true eigenmode weight. The zone-boundary minimum is at L; red marks the shallow soft valley along $(t, \frac{1}{2}, \frac{1}{2})$ (Sec. XII A). (b) Ground-state energy per site against $1/N$: exact diagonalization (filled) and Green's-function Monte Carlo (open, with error bars) on one axis. At the two sizes where both methods exist, 48 and 72 sites, they agree within statistical errors — the method gate, shown explicitly. (c) The ratio at L and X across both methods and all sizes: the zone-boundary hierarchy, L softest, holds at every size, while the interior of the zone (not shown) drifts strongly — the reason large systems were required.

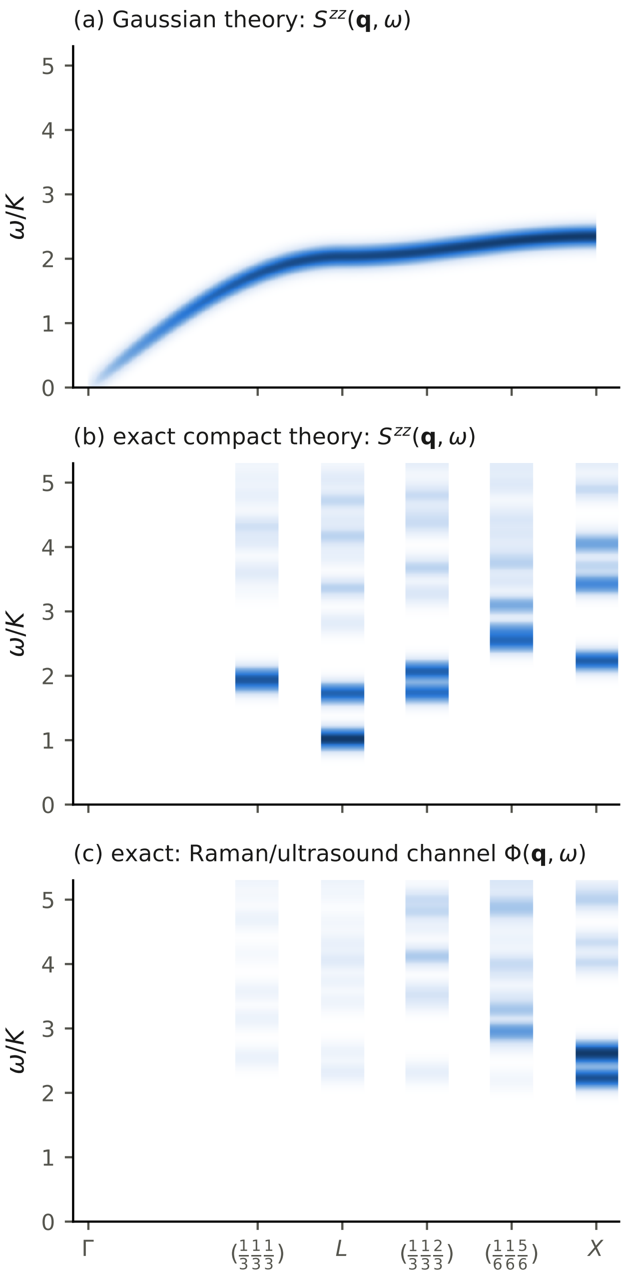

FIG. 5. Simulated spectra along Γ–L–X for the 48-site cluster, broadened by $\eta = 0.08K$ and normalized separately in each panel. (a) Gaussian one-photon neutron response. (b) Exact operator-resolved S^z response within the ground-state winding sector: split photon descendants, a dominant low L mode at about half the Gaussian-photon energy, and higher off-pole weight. (c) Spin-reversal-even response accessible to bilinear probes. By the parity selection rule no eigenstate is shared between panels (b) and (c).